Sensing the forest through the trees

A data driven approach to Dutch forest reserve monitoring using AHN

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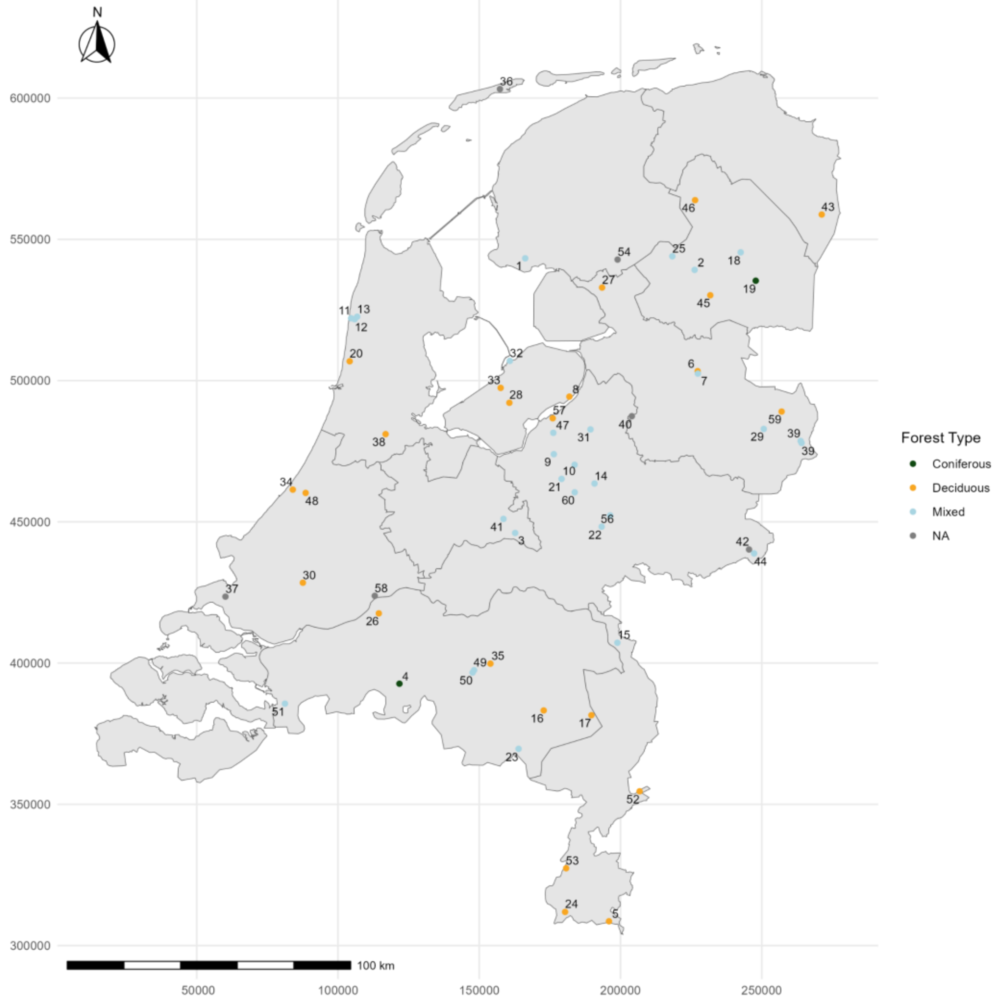
Contents

Introduction – The Dutch forest reserves I

- Established in 1987
- To follow natural development of Dutch forests
- Little knowledge, virtually no natural forests left
- 59 reserves
- 11 of 12 provinces, ~2600 hectares
- 10-yearly monitoring cycle
- Budget cuts 2005
- Ad-hoc basis some monitoring, national scale has ceased
- Looking to start monitoring again

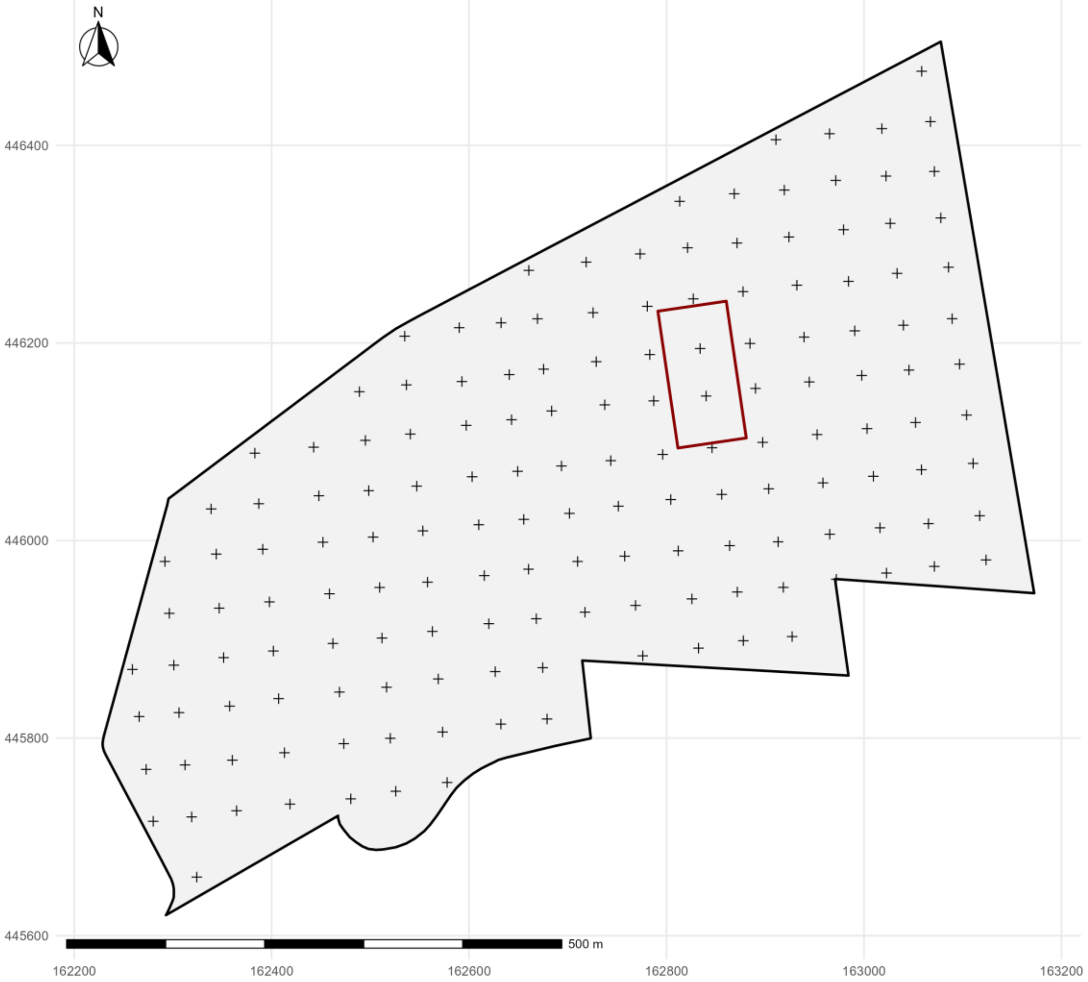


Introduction – The Dutch forest reserves II

- Different measurement levels
  - Core area
  - Sample sites
- Different variables
- Time intensive & expensive

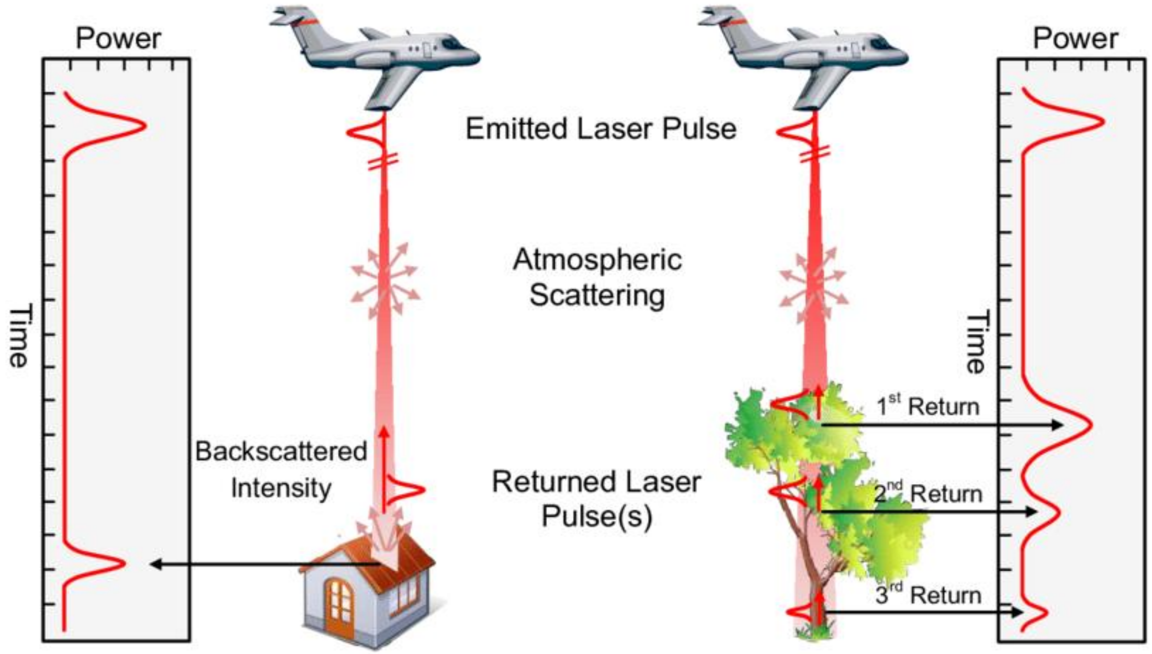
Table 1: Overview of forest reserve characteristics monitored at the three unique monitoring scales. dbh = diameter at breast height. h = height.

Variable	Transect	Core Area	Sample Plots
Plant species distribution	x	x	x
Cover percentage of plants	x	x	x
Shrub presence (dbh<5cm, h<0.5m)	x		
Shrub presence (dbh<5cm, h>0.5m)		x	x
Tree and shrub seedlings		x	x
Tree presence (dbh>5cm)		x	x
Tree position		x	x
Tree crown projection		x	
Tree diameter at breast height		x	x
Tree height		x	x
Tree crown characteristics		x	x
Tree vitality		x	x
Damage to trees		x	x
Decomposition stage of dead trees		x	x

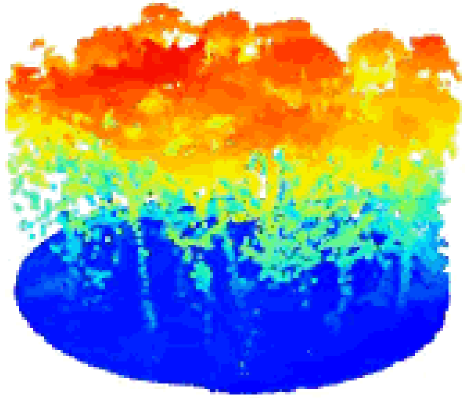


Introduction – LiDAR

- Active remote sensing technique
- Send pulses with a sensor in Near Infrared
- Measures when it “hits” something
- Can be a single hit, or multiple hit
- Allow 3D measurements of objects, areas, forests etc.
- Particularly interesting for trees, as these often have multiple hits
- TLS, MLS, ALS, UAV-based

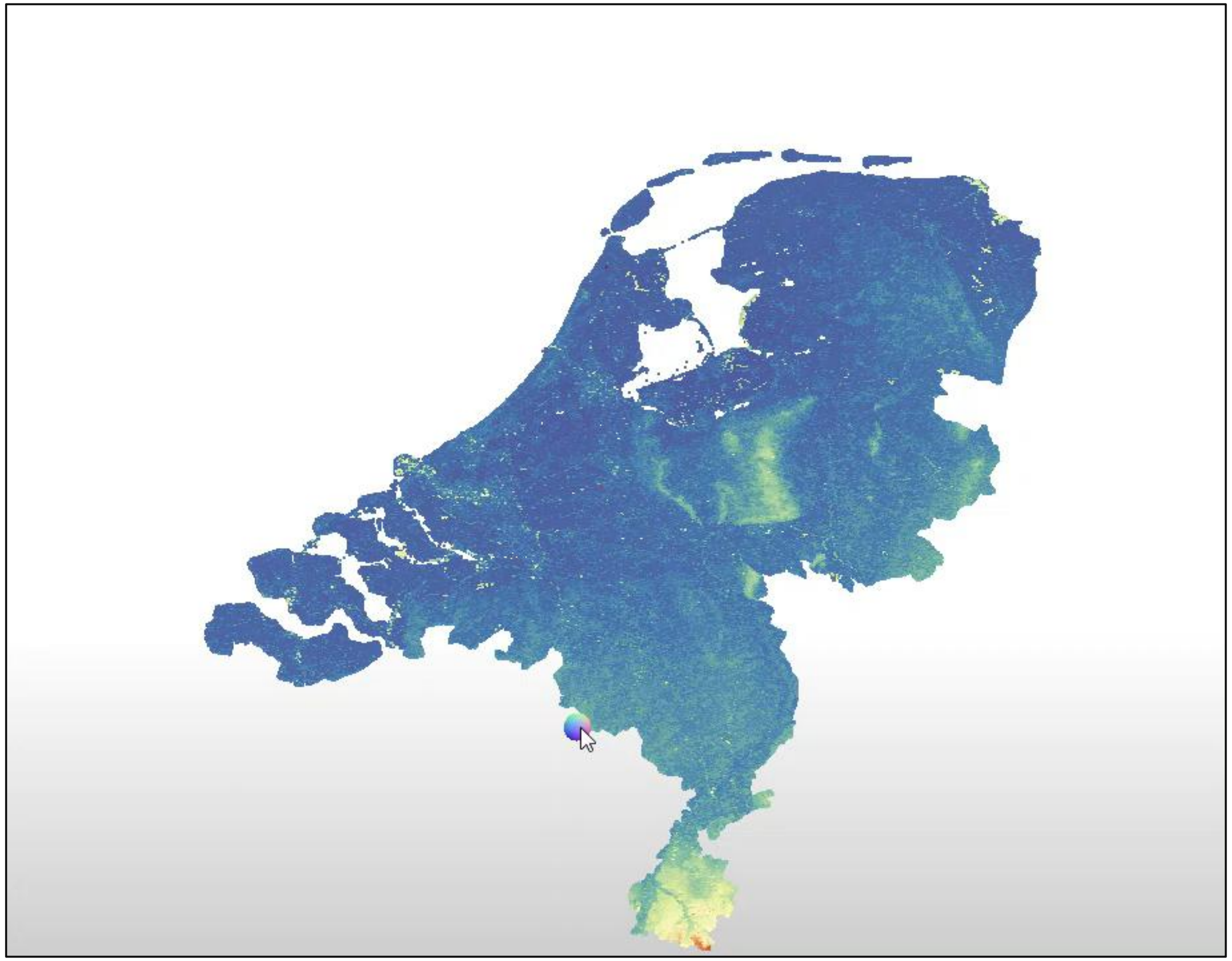


Yan, W. Y., Shaker, A., & El-Ashmawy, N. (2014). Urban land cover classification using airborne LiDAR data: A review. *Remote Sensing of Environment*, *158*, 295–310. https://doi.org/10.1016/j.rse.2014.11.001



Introduction – AHN

- Nationwide Dutch ALS
- Different products
  - Rasters
  - Raw point clouds
- Different versions – AHN1 to AHN6
- Correspond with different times
- Different quality levels
- Open access



Research Goals

- Currently AHN is not used much in the forestry sector
- In theory, AHN might be able to contribute to the forest reserve monitoring
- Is this true in practice?

1: Identify and evaluate potential methods utilising AHN for analysing the forest reserve characteristics

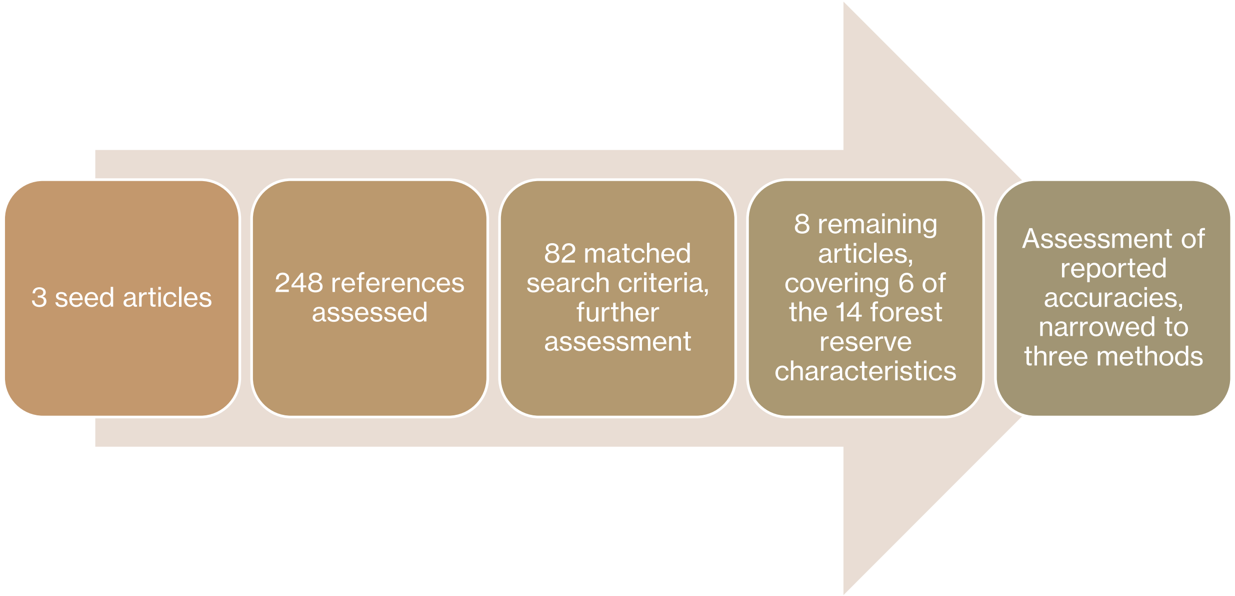
2: Assess the feasibility of applying the found methods to AHN data

3: Evaluate the suitability of AHN-applied methods

Literature Review

1: Identify and evaluate potential methods utilising AHN for analysing the forest reserve characteristics

- Brief literature review
- Single iteration backward snowballing
- Three seed articles
- All references’ abstracts assessed against four search criteria
- Those passed read in detail, assessed against two more criteria

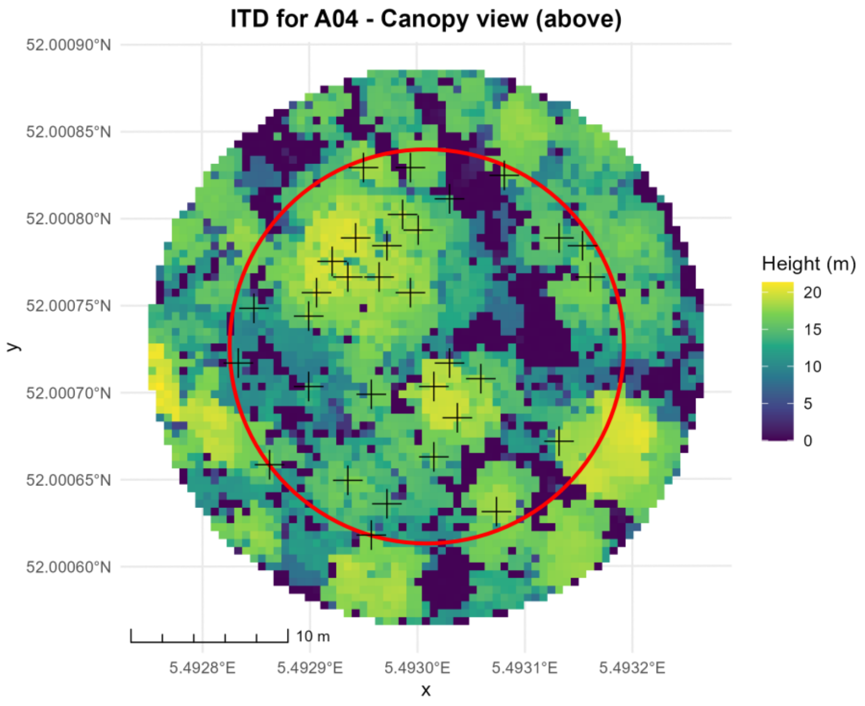


ITD - 1

2: Assess the feasibility of applying the found methods to AHN data

- Individual Tree Detection
- Accurately detecting the location of individual trees and their height
- Method by Sparks et al. (2022)

LiDAR Point Cloud → Canopy Height Model (50cm) → Locate treetops with a 1.5m radius Local Maxima Filter



ITD – 2

2: Assess the feasibility of applying the found methods to AHN data

- Assessed the ITD of 9 sample sites from three representative reserves
- Saw overclassification generally
- Height of the trees generally good results
- Conifers performed a bit better (likely due to conical shape)
- How to improve?
- Variable Window Size?

Table 6: Confusion matrix constructed based on visual assessment of individual tree detection

	True positives (TP)	False positives (FP)	False negatives (FN)
RSV 3: Mixed forest	43	32	4
RSV 16: Deciduous forest	48	48	9
RSV 19: Coniferous forest	41	21	5

Table 3: Derived linear equations for Variable Window Size approach to Individual Tree Detection in SRO3.

Forest type	**Equation (r = radius (m), h = height (m))**
Mixed	$r = -0.856 + 0.178h$
Deciduous	$r = 1.786 + 0.056h$
Coniferous	$r = -0.843 + 0.145h$

ITD – 3

3: Evaluate the suitability of AHN-applied methods

- Check results against field data from field work in 2018
- Try to match the locations between field-observed trees and ITD-derived trees
- Low accuracy
- 5.7% detection rate
- Not suitable for ITD
- But for height, valuable

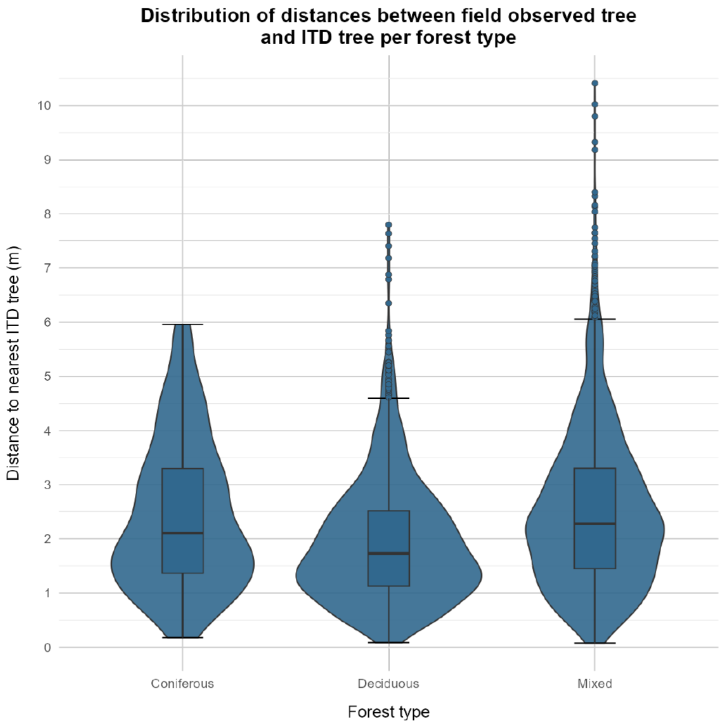


Figure 9: Comparison and distribution of the minimal distance between trees identified in the field and through ITD.

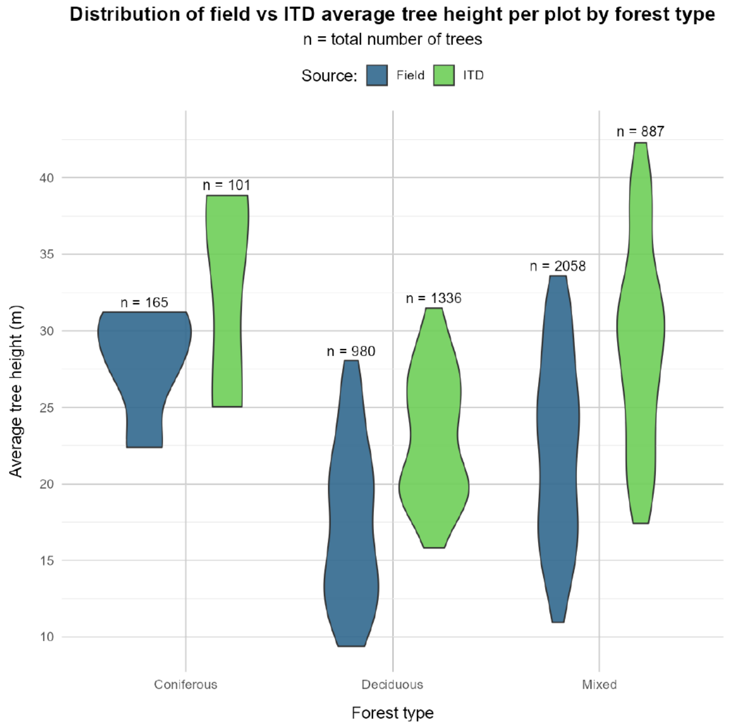


Figure 10: Distribution of average tree heights in plots across forest types.

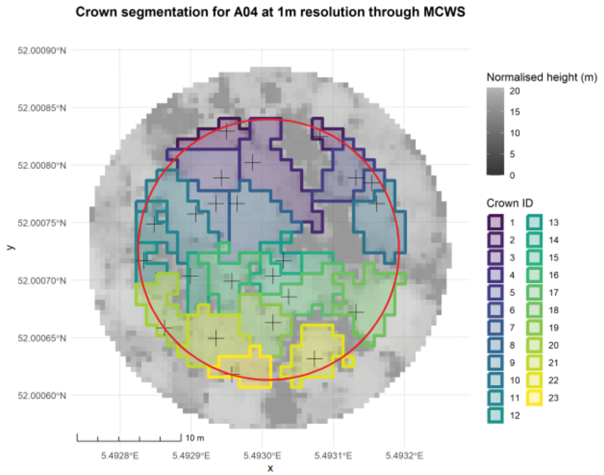
Crown projection - 1

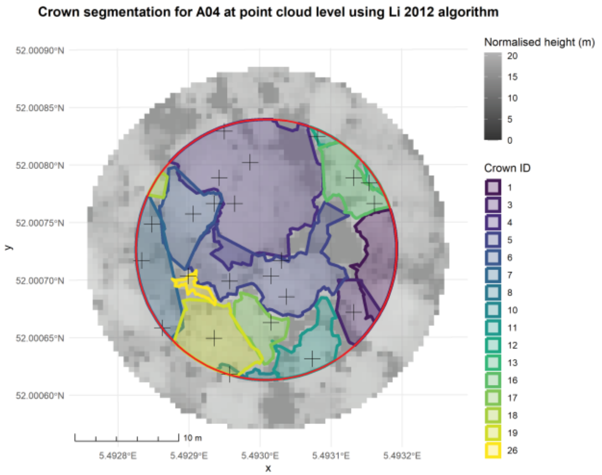
2: Assess the feasibility of applying the found methods to AHN data

- Determination of the shape of the crown projection
- Method by Bazezew et al. (2018)
- Not possible, uses eCognition, paid software
- Instead, two open source alternatives
- First, Multicore Watershed Segmentation with a pit-free CHM at intervals 0, 2, 5, 10 and 15m, with ITD input
- Second, point cloud based 3D segmentation, Li 2012 algorithm

Table 7: Confusion matrix constructed based on visual assessment of tree crown projection

	MCWS		Li et al., 2012		Reference
	Correct (n_c)	Incorrect (n_i)	Correct (n_c)	Incorrect (n_i)	(n_r)
RSV 3: Mixed	34	39	37	17	47
RSV 16: Deciduous	39	56	46	21	64
RSV 19: Coniferous	31	29	47	16	62





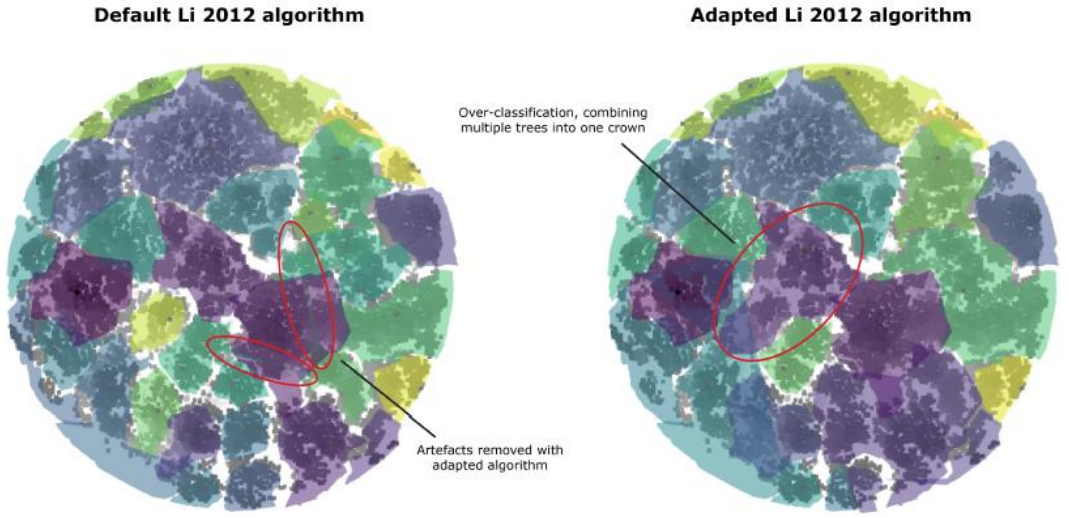
Crown projection - 2

3: Evaluate the suitability of AHN-applied methods

- Comparison to field work not possible
- Parameter finetuning
- Generally, a trade-off
- No significant improvements with the different parameter combinations
- Suitable for application to AHN?

Table 8: Results of visual comparison of different parameter combinations in the Li 2012 algorithm against the default parameters.

$dt1$	$dt2$	Z_u	h_{min}	Visual results
1.5	2.00	15	4	No notable differences
1.5	2.00	10	2	No notable differences
1.5	2.00	20	2	No notable differences
1.5	2.00	30	2	No notable differences
1.5	3.00	15	2	Over-classification
2.0	3.00	15	2	Over-classification
1.0	2.00	15	2	No notable differences
1.0	1.50	15	2	Over-segmentation
1.5	1.75	25	2	Minimal over-segmentation
1.6	1.85	20	5	Minimal over-segmentation
1.7	2.00	20	4	No notable differences

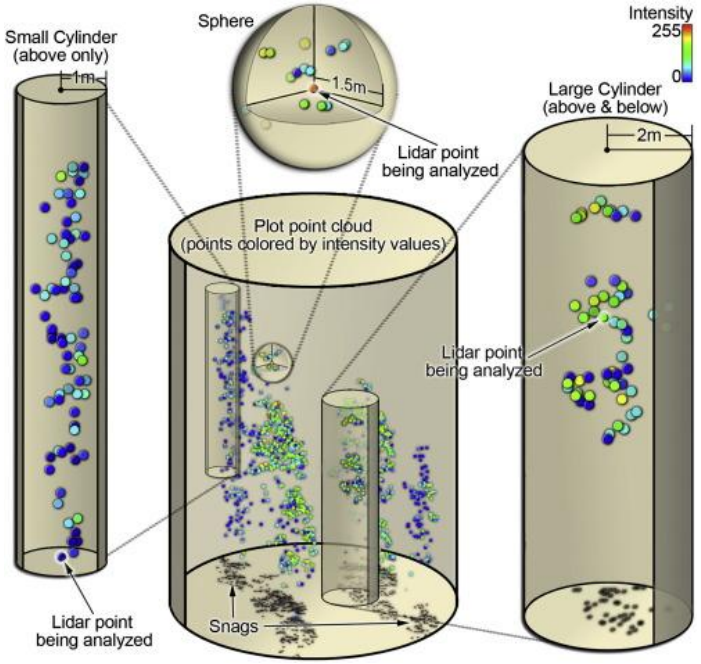


(a) Comparison of default and adapted Li 2012 algorithm for sample plot S09 in forest reserve *Galgenberg* (RSV 3)

Snag detection - 1

2: Assess the feasibility of applying the found methods to AHN data

- Detection of standing dead trees
- Method by Wing et al. (2015)
- Intensity values of the LiDAR points, assumption that the intensity patterns of live tree and dead tree points are different
- Calculate plot-level variables; point density, maximum intensity index, fraction of canopy cover, mean canopy height and the Branch and Bole versus Foliage ratio
- Neighbourhood attribute filtering in cylinders and sphere



Wing, B. M., Ritchie, M. W., Boston, K., Cohen, W. B., & Olsen, M. J. (2015). Individual snag detection using neighborhood attribute filtered airborne lidar data. *Remote Sensing of Environment*, *163*, 165–179. https://doi.org/10.1016/j.rse.2015.03.013

Snag detection - 2

2: Assess the feasibility of applying the found methods to AHN data

- Over classification, ~90% of points classed as snag points
- Unrealistically high
- Made changes to the BBPR matrix
- No effects, still overclassification
- Not feasible
- What is the problem?

Table2. The four groups of conditional assessments with their required neighborhood point density and average BBPR values (BBPR=branch and bole point ratio; PDR=point density requirement). Values are associated with an algorithm sensitivity level of 4.

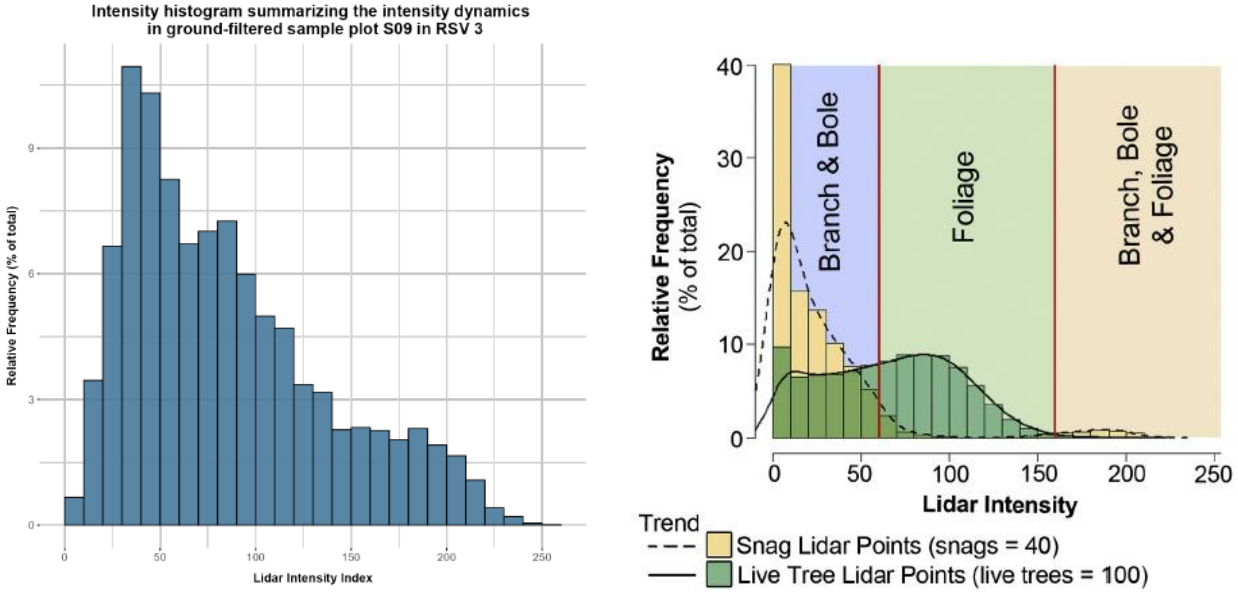
Assessment group	Sphere		Small cylinder		Large cylinder	
	Point density (>=)	Average BBPR (>=)	Point density (>=)	Average BBPR (>=)	Point density (>=)	Average BBPR (>=)
General	PDR	0.99	–	0.99	–	0.70
	PDR	0.95	–	0.95	–	0.725
	PDR	0.90	–	0.90	–	0.75
	PDR	0.85	–	0.85	–	0.775
	PDR	0.80	–	0.80	–	0.80
Small snag	2 and (<=) PDR	0.95	2 and (<=) PDR	0.95	2 and (<=) PDR	0.60
	2 and (<=) PDR	0.90	2 and (<=) PDR	0.90	2 and (<=) PDR	0.65
	2 and (<=) PDR	0.85	2 and (<=) PDR	0.85	2 and (<=) PDR	0.75
Live crown edge	PDR	0.80	PDR	0.95	PDR*7	0.70
	PDR	0.85	PDR	0.90	PDR∗7	0.75
	PDR	0.90	PDR	0.85	PDR∗7	0.80
	PDR	0.95	PDR	0.80	PDR∗7	0.85
High canopy cover (only CC>=55%)	PDR	0.95	PDR	0.95	PDR∗8	0.75
	…	…	…	…	…	
	PDR	0.95	PDR	0.95	PDR∗15	0.55
	PDR	0.90	PDR	0.90	PDR∗8	0.85
	…	…	…	…	…	…
	PDR	0.90	PDR	0.90	PDR∗15	0.65

PDR: Point density requirement based on plot-level point density (PLPD) defined classes.

Snag detection - 3

3: Evaluate the suitability of AHN-applied methods

- Explored differences between the datasets
  - Different sensor
  - Different intensity storage
  - Scan angles
  - Flying season
  - Study area
  - Single and multiple returns
  - Ground filtering
  - Point density
- Changes had little effect
- Fundamental mismatch
- Suitable?



(a) Distribution of intensity index across representative sample sites

(b) Distribution of intensity index presented by Wing et al. (2015)

Figure 12: Comparison of distributions of intensity index between representative sample sites and the article presented by Wing et al. (2015)

Discussion
General

A lot of knowledge available

Literature review too constrained and too wide

- Time cutoff
- Forest type
- Only reverse snowballing

From literature to methods, not as easy as it seems

Discussion
ITD

Original study coniferous forests

- Works better with circular FWS

Relationships between height and crown width

- Linear is not most realistic, ideally quadratic

CHMs lead to data loss

- Forcing 3D data to be 2D
- Li (2012) algorithm?

Field work data the truth?

Discussion
Crown projection

- MCWS outperformed by Li 2012
- Detects canopy cover well, linking to trees more difficult
- Even identifies overlapping crowns on different heights
- Comparison to field data needed

Discussion
Standing dead tree detection

- Poor results
- Fundamental mismatch
- Too many environmental variables
- Unlikely to work out of the box

Discussion
Closing

- Individual versus area-based
- Time to build models
- AHN6 and beyond
- Replacement or support tool

Conclusions

1: There are vast amounts of methods available, though practical application is difficult and time consuming

2: ITD and crown projection showed feasibility, snag detection is unfeasible for AHN data.

3: Crown projection is suitable in current workflow, as is height from ITD. Much more that not yet

Looking forward

- Lots of potential research
  - Additional methods
  - Area-based approaches
  - Machine / Deep Learning methods training
- AHN6 introduces new approaches
- To start, ask the fundamental questions and build on that
- Then, more specialised literature review
- Build on that

Questions?